METRIC

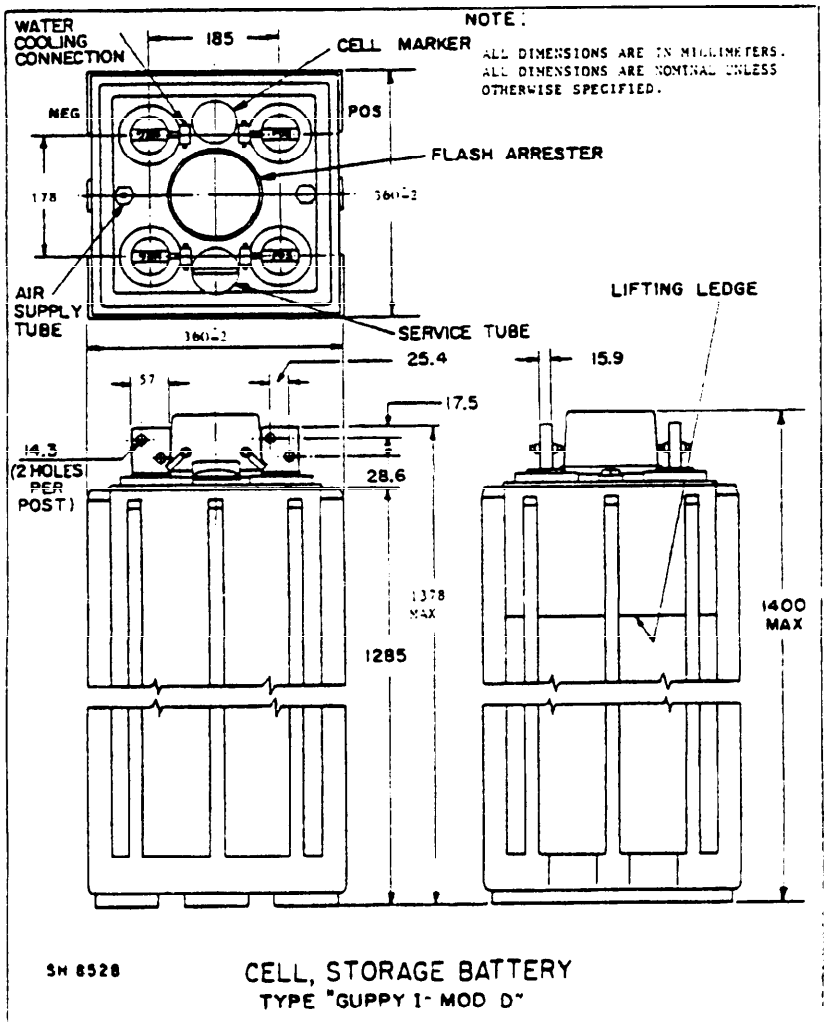
DOD-B-24541/4(SH)
24 November 1982

MILITARY SPECIFICATION SHEET

BATTERY CELLS AND ELEMENTS, LEAD-ACID, MAIN STORAGE, SUBMARINE, TYPE GUPPY I, MOD D (METRIC)

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery cells and elements described herein shall consist of this document and the latest issue of DOD-B-24541.



FSC 6140

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REQUIREMENTS:

Electrolyte specific gravity: 1.285 nominal.
Measurement of capacity: Capacity, in ampere-hours (Ah), is based on an initial electrolyte temperature of 26.7 degrees Celsius, specific gravity of 1.285, and shall be not less than as specified in table I.
Water cooling: Required.

TABLE I. Electrical characteristics.

Capacity Ah	Time rate in hours	Discharge current A	Minimum battery voltage				Minimum cell voltage	
			126-cell group		252-cell group			
			Average	Final	Average	Final	Average	Final
4000	1	4000	207	180	414	360	1.46	1.28
5505	3	1835	228	201	456	402	1.61	1.51
6100	5	1220	236	207	472	414	1.65	1.57
6270	6	1045	238	208	476	416	1.66	1.58
6850	10	685	241	212	482	424	1.69	1.63
7400	20	370	244	215	488	430	1.71	1.66
7740	36	215	246	217	492	434	1.72	1.68
7920	48	165	247	217	494	434	1.72	1.68

Mass per cell (filled and charged): Maximum - 502 kilograms (kg)
Maximum average - 493 kg
Minimum - 470 kg

Guaranteed service life: (equivalent cycle equals 5,000 Ah of charge) 300 equivalent cycles or 30 months, whichever occurs first.

Demonstration of rated capacity: Within 30 equivalent cycles.
Life test: Cycling regime.

Hydrogen emission (maximum): (measurements recorded as cubic centimeters/hour/cell/1,000 Ah at the 10-hour rate)
5,500 at 200 A
3,500 at 100 A
100 on open circuit

NOTE: Positive plate thickness shall be not less than 4.8 millimeters.

Preparing activity:
Navy - SH
(Project 6140-N564)

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